

SECOND-ORDER LINEAR HOMOGENEOUS RECURRENCE RELATIONS

Consider a method of solving recurrence relations of the form $a_n + c_1 a_{n-1} + c_2 a_{n-2} = 0$, $n \geq 2$ (or) $C_n a_n + C_{n-1} a_{n-1} + C_{n-2} a_{n-2} = 0$, $n \geq 2$, where C_n, C_{n-1}, C_{n-2} are ⁽¹⁾ real constants with $C_n \neq 0$. This is known as 2nd order recurrence relations with constant coefficients.

Put $a_n = a^n$, $a_{n-1} = a^{n-1}$, $a_{n-2} = a^{n-2}$.

$\therefore a^n + c_1 a^{n-1} + c_2 a^{n-2} = 0 \rightarrow (2)$
Equation (2) is called an auxiliary equation (or) the characteristic equation for the relation (1).

METHOD OF CHARACTERISTIC ROOTS:

- ① If the two roots are real and different then $a_n = AK_1^n + BK_2^n$.
- ② If the two roots are equal (say $K_1 = K_2$) then $a_n = (A + Bn)K^n$.
- ③ If the roots are complex then $a_n = r^n (A \cos n\theta + B \sin n\theta)$ where A & B are constants $r = |K_1| = |K_2| = \sqrt{p^2 + q^2}$ and $\theta = \tan^{-1}(q/p)$.

① Solve the recurrence relation $f_n = f_{n-1} + f_{n-2}$ with $f_0 = 0$, $f_1 = 1$.

Sol: Given $f(n) = f(n-1) + f(n-2)$
The char. eqn.
put $f(n) = a^n$, $f(n-1) = a^{n-1}$, $f(n-2) = a^{n-2}$.

Now $a^n = a^{n-1} + a^{n-2}$

$a^n - a^{n-1} - a^{n-2} = 0$.

put $n=2$, $a^2 - a - 1 = 0$.

$$a = \frac{-(-1) \pm \sqrt{(-1)^2 - 4 \times 1 \times -1}}{2 \times 1}$$

$$= \frac{1 \pm \sqrt{1+4}}{2}$$

$$= \frac{1 \pm \sqrt{5}}{2}$$

$$K_1 = a = \frac{1+\sqrt{5}}{2}, \quad K_2 = a = \frac{1-\sqrt{5}}{2}$$

[If $ax^2 + bx + c = 0$ then $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$]

$\therefore$ The roots are

$$a_n = AK_1^n + BK_2^n$$

$$a_n = A\left(\frac{1+\sqrt{5}}{2}\right)^n + B\left(\frac{1-\sqrt{5}}{2}\right)^n$$

③

$$f_n = a_n = A \left(\frac{1+\sqrt{5}}{2} \right)^n + B \left(\frac{1-\sqrt{5}}{2} \right)^n$$

Sol

Put $n=0$, $a_0 = A+B$

$$f(0) = A+B$$

$$0 = A+B \rightarrow (1)$$

Put $n=1$, $a_1 = A \left(\frac{1+\sqrt{5}}{2} \right)^1 + B \left(\frac{1-\sqrt{5}}{2} \right)^1$

$$f(1) = A \left(\frac{1+\sqrt{5}}{2} \right) + B \left(\frac{1-\sqrt{5}}{2} \right)$$

$$1 = A \left(\frac{1+\sqrt{5}}{2} \right) + B \left(\frac{1-\sqrt{5}}{2} \right)$$

$$2 = A(1+\sqrt{5}) + B(1-\sqrt{5}) \rightarrow (2)$$

$$(1) \Rightarrow \boxed{A = -B}$$

$$2 = -B(1+\sqrt{5}) + B(1-\sqrt{5})$$

$$= -B - B\sqrt{5} + B - B\sqrt{5}$$

$$2 = -2B\sqrt{5}$$

$$\boxed{-\frac{1}{\sqrt{5}} = B}, \quad \boxed{A = \frac{1}{\sqrt{5}}}$$

The General solution is

$$f_n = a_n = \left(\frac{1}{\sqrt{5}} \right) \left(\frac{1+\sqrt{5}}{2} \right)^n + \left(-\frac{1}{\sqrt{5}} \right) \left(\frac{1-\sqrt{5}}{2} \right)^n$$

② Solve the recurrence relation $a_n = -3a_{n-1} - 3a_{n-2} - a_{n-3}$ with initial condition $a_0=5, a_1=9, a_2=15$.

Sol: Given $a_n + 3a_{n-1} + 3a_{n-2} + a_{n-3} = 0$.

The char. eqn $a^n + 3a^{n-1} + 3a^{n-2} + a^{n-3} = 0$.

Take $a_n = a^n$ & so on.

Put $n=3$, $a^3 + 3a^2 + 3a + 1 = 0$.

$\therefore a = -1$, is a root.

$$-1 \left| \begin{array}{cccc} 1 & 3 & 3 & 1 \\ 0 & -1 & -2 & -1 \\ \hline 1 & 2 & 1 & 0 \end{array} \right|$$

$$a^2 + 2a + 1 = 0$$

$$(a+1)(a+1) = 0$$

$$a = -1, a = -1$$

$\therefore$ The roots are $-1, -1, -1$.

$$a_n = (A + Bn + Cn^2) K^n = (A + Bn + Cn^2) (-1)^n \rightarrow (1)$$

Given $a_0 = 5, a_1 = 9, a_2 = 15$.

Put $n=0$ in (1)

$$a_0 = (A + B(0) + C(0)) (-1)^0 = (A + 0 + 0)(1) = A$$

$$\boxed{5 = A}$$

Put $n=1$, in equ (1)

$$a_1 = (A + B + C) (-1) = -A - B - C$$

$$9 = -5 - B - C$$

$$14 = -B - C$$

$$B + C = -14 \rightarrow (2)$$

Put $n=2$ in equ (1)

$$a_2 = (A + 2B + 4C) (-1)^2$$

$$15 = A + 2B + 4C$$

$$15 = 5 + 2B + 4C$$

$$10 = 2B + 4C$$

$$5 = B + 2C \rightarrow (3)$$

Solve (2) & (3)

$$B + C = -14$$

$$B + 2C = 5$$

$$\begin{array}{r} (-) \quad (-) \quad (-) \\ \hline -C = -19 \end{array}$$

$$\boxed{C = 19}$$

$$\begin{array}{r} (x) \downarrow \\ 1 \quad 2 \\ \hline (+) 2 \end{array}$$

(2)

$$\begin{aligned} (3) \Rightarrow 5 &= B + 2(19) \\ 5 &= B + 38 \end{aligned}$$

$$\boxed{-33 = B}$$

$$\therefore A = 5, B = -33, C = 19$$

The General solution is

$$a_n = (5 - 33n + 19n^2) (-1)^n$$

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③ Solve the R.R $a_n - 3a_{n-1} + 2a_{n-2} = 0$, for $n \geq 2$ given that $a_0 = 5$ and $a_1 = 3$.

Sol: Given $a_n - 3a_{n-1} + 2a_{n-2} = 0$

We can write the equ $a^n - 3a^{n-1} + 2a^{n-2} = 0$.

Put the char equ

$$n=2, \quad a^2 - 3a + 2 = 0$$

$$(a-2)(a-1) = 0.$$

$$\boxed{a=2} \quad \boxed{a=1}$$

$$\begin{array}{r} 2 \\ (-) \quad 2 \quad \swarrow \quad \searrow \quad 1 \\ (+) \quad -3 \end{array}$$

$$\therefore a_n = AK_1^n + BK_2^n = A(2)^n + B(1)^n$$

Put $n=0$, $a_0 = A(2)^0 + B(1)^0$

$$5 = A + B \rightarrow (1)$$

Put $n=1$, $a_1 = A(2)^1 + B(1)^1$

$$3 = 2A + B \rightarrow (2)$$

Solve (1) & (2), we get

$$A + B = 5$$

$$2A + B = 3$$

$$\begin{array}{r} (-) \quad (-) \quad (-) \\ \hline -A = 2 \end{array}$$

$$\boxed{A = -2}$$

$$\textcircled{1} \Rightarrow A + B = 5$$

$$-2 + B = 5$$

$$B = 5 + 2 = 7$$

$$\therefore \boxed{B = 7}$$

The General solution is

$$a_n = A(2)^n + B(1)^n = (-2)(2)^n + 7(1)^n.$$

④ Solve the R.R $s(n) = s(n-1) + 2s(n-2)$ given $s(0) = 3$, $s(1) = 1$.

Sol: Given $s(n) - s(n-1) - 2s(n-2) = 0$.

We can write the above equ $a^n - a^{n-1} - 2a^{n-2} = 0$ | take $s(n) = a^n$ & so on.

$$a^n - a^{n-1} - 2a^{n-2} = 0.$$

Put $n=2$, $a^2 - a - 2 = 0$

$$(a-2)(a+1) = 0.$$

$$a = 2, a = -1.$$

The char roots are

$$a_n = AK_1^n + BK_2^n = A(2)^n + B(-1)^n.$$

Given $s(0) = 3$, $s(1) = 1$.

Put $n=0$, $a_0 = A(2)^0 + B(-1)^0$

$$3 = A + B \rightarrow (1)$$

Put $n=1$, $a_1 = A(2)^1 + B(-1)^1$

$$1 = 2A - B \rightarrow (2)$$

Solve (1) & (2)

$$A + B = 3$$

$$2A - B = 1$$

$$\hline 3A = 4$$

$$\boxed{A = \frac{4}{3}}$$

$$\textcircled{1} \Rightarrow A + B = 3$$

$$\frac{4}{3} + B = 3$$

$$B = 3 - \frac{4}{3} = \frac{9-4}{3} = \frac{5}{3}$$

$$\boxed{B = \frac{5}{3}}$$

The general solution is

$$a_n = \left(\frac{4}{3}\right)(2)^n + \left(\frac{5}{3}\right)(-1)^n.$$

⑤ solve the R.R $S_n + 3S_{n-1} - 4S_{n-2} = 0$ given that $a_0 = 2$, $a_1 = 8$.

Sol: Given $a^n + 3a^{n-1} - 4a^{n-2} = 0$

Put $n=2$, $a^2 + 3a - 4 = 0$.

$$\begin{array}{r} (x) \quad 2 \\ -2 \quad \diagup \quad 1 \\ \hline (+) \quad -1 \end{array}$$

⑤

Take
 $a_n = S_n = a^n$
& so on.

$$(a+4)(a-1)=0$$

$$a = -4, a = 1.$$

The char roots are

$$a_n = AK_1^n + BK_2^n$$

$$a_n = A(-4)^n + B(1)^n.$$

Put $n=0$, $a_0 = A(-4)^0 + B(1)^0$

$$2 = A+B \rightarrow (1)$$

Put $n=1$, $a_1 = A(-4)^1 + B(1)^1$

$$8 = -4A+B \rightarrow (2)$$

Solve (1) & (2), $A+B=2$

$$-4A+B=8$$

$$\begin{array}{r} (+) \quad (-) \quad (-) \\ 5A = -6 \end{array}$$

$$\boxed{A = -\frac{6}{5}}$$

$$(1) \Rightarrow A+B=2$$

$$-\frac{6}{5} + B = 2$$

$$B = 2 + \frac{6}{5}$$

$$\boxed{B = \frac{16}{5}}$$

The General sol is

$$a_n = \left(-\frac{6}{5}\right)(-4)^n + \left(\frac{16}{5}\right)(1)^n.$$

⑥ Solve the R.R $a_n + a_{n-1} - 6a_{n-2} = 0$, $n \geq 2$ given that $a_0 = -1$, $a_1 = 8$.

Sol: Given $a^n + a^{n-1} - 6a^{n-2} = 0$,

The char equ

Put $n=2$

$$a^2 + a - 6 = 0$$

$$(a+3)(a-2) = 0$$

$$a = -3, a = 2.$$

The char roots are $a_n = AK_1^n + BK_2^n = A(-3)^n + B(2)^n$

at $n=0$, $a_0 = A(-3)^0 + B(2)^0$

$$-1 = A+B \rightarrow (1)$$

at $n=1$, $a_1 = A(-3)^1 + B(2)^1$

$$8 = -3A + 2B \rightarrow (2)$$

$(1) \times (2)$

$$(1) \times 2 \Rightarrow 2A + 2B = -2$$

$$(2) \times 1 \Rightarrow -3A + 2B = 8$$

$$\begin{array}{r} (+) \quad (-) \quad (-) \\ 5A = -10 \end{array}$$

$$\boxed{A = -2}$$

$$(1) \Rightarrow A+B = -1$$

$$-2 + B = -1$$

$$B = -1 + 2 = 1$$

$$\boxed{B = 1}$$

The General solution is

$$a_n = (-2)(-3)^n + (1)(2)^n.$$

⑦ Solve the R.R $a_n - 6a_{n-1} + 9a_{n-2} = 0$, $n \geq 2$ given that $a_0 = 5$ ⑦

$$a_1 = 12.$$

Sol:- Given $a^n - 6a^{n-1} + 9a^{n-2} = 0$

The char equ

$$\text{Put } n=2, a^2 - 6a + 9 = 0$$

$$(a-3)(a-3) = 0$$

$$a = 3, a = 3$$

The char roots are $a_n = (A+Bn)K^n = (A+Bn)3^n$.

$$\text{Put } n=0, a_0 = (A+B(0))3^0 = (A+0)(1) = A$$

$$\boxed{5 = A}$$

$$\text{Put } n=1, a_1 = (A+B)3^1 = 3A + 3B$$

$$12 = 3A + 3B$$

$$4 = A + B$$

$$4 = 5 + B$$

$$4 - 5 = B \Rightarrow \boxed{B = -1}$$

$$\text{The General sol is } a_n = (A+Bn)3^n = (5-n)3^n.$$

⑧ Solve the R.R $a_n = 2(a_{n-1} - a_{n-2})$, $n \geq 2$ given that $a_0 = 1, a_1 = 2$.

Sol:- Given $a_n = 2(a_{n-1} - a_{n-2})$

$$a_n = 2a_{n-1} - 2a_{n-2}$$

The equ can write as $a^n - 2a^{n-1} + 2a^{n-2} = 0$

The char equ

$$\text{Put } n=2,$$

$$a^2 - 2a + 2 = 0$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-(-2) \pm \sqrt{(-2)^2 - 4 \times 1 \times 2}}{2 \times 1}$$

$$= 2 \pm \sqrt{4-8}$$

$$a = \frac{2 \pm \sqrt{-4}}{2}$$

$$= \frac{2 \pm i2}{2} \quad (p \pm iq)$$

$$= 1 \pm i \quad (\alpha \pm i\beta)$$

$$\alpha = 1, \beta = 1$$

The char roots are $(p \pm iq)$
 $(\alpha \pm i\beta)$

$$a_n = r^n (A \cos n\theta + B \sin n\theta)$$

$$\text{where } r = \sqrt{p^2 + q^2}, \theta = \tan^{-1}(q/p)$$

$$= \sqrt{1+1}$$

$$\boxed{r = \sqrt{2}}$$

$$\theta = \tan^{-1}(1)$$

$$\boxed{\theta = \frac{\pi}{4}}$$

Hence

$$a_n = (\sqrt{2})^n [A \cos \frac{n\pi}{4} + B \sin \frac{n\pi}{4}]$$

Put $n=0$, $a_0 = (\sqrt{2})^0 [A \cos 0 + B \sin 0] = A + 0$

$\boxed{1 = A}$

Put $n=1$, $a_1 = (\sqrt{2})^1 [A \cos \frac{\pi}{4} + B \sin \frac{\pi}{4}]$

$2 = (\sqrt{2}) \left[\frac{A}{\sqrt{2}} + \frac{B}{\sqrt{2}} \right]$

$\frac{2\sqrt{2}}{\sqrt{2}} = A + B$

$A + B = 2$

$1 + B = 2$

$B = 2 - 1 = 1$

$\boxed{B = 1}$

Hence the General sol is

$a_n = (\sqrt{2})^n \left[1 \cos \frac{n\pi}{4} + 1 \sin \frac{n\pi}{4} \right]$

Q) Solve the R.R $2a_{n+3} = a_{n+2} + 2a_{n+1} - a_n$ for $n \geq 0$ given that

$a_0 = 0, a_1 = 1, a_2 = 2.$

Sol:- Given

$2a^{n+3} - a^{n+2} - 2a^{n+1} + a^n = 0.$

Put $n=0$,

$2a^3 - a^2 - 2a + 1 = 0.$

$a^2(2a-1) - 1(2a-1) = 0$

$(a^2-1)(2a-1) = 0$

$a^2 - 1 = 0 \quad | \quad 2a - 1 = 0$

$a^2 = 1$

$2a = 1$

$\boxed{a = \frac{1}{2}}$

$\boxed{a = \pm 1}$

The char roots are

$a_n = AK_1^n + BK_2^n + CK_3^n$

$a_n = A(1)^n + B(-1)^n + C\left(\frac{1}{2}\right)^n.$

Put $n=0$, $a_0 = A + B + C \rightarrow (1)$

Put $n=1$, $a_1 = A - B + \frac{1}{2}C$

$1 = A - B + \frac{C}{2} \rightarrow (2)$

Solve (1) & (2)

$A + B + C = 0$

$A - B + \frac{C}{2} = 1$

$\frac{2A}{2} + \frac{3C}{2} = 1 \rightarrow (3)$

Put $n=2$,

$a_2 = A(1)^2 + B(-1)^2 + C\left(\frac{1}{2}\right)^2$

$2 = A - B + \frac{1}{4}C \rightarrow (4)$

Solve (2) & (4)

$A - B + \frac{C}{4} = 2$

$A - B + \frac{C}{2} = 1$

$(-)(+)(-)(-)$

$\frac{C}{4} - \frac{C}{2} = 1$

$-\frac{C}{4} = 1$

$\boxed{C = -\frac{1}{4}}$

$\boxed{A = \frac{11}{16}}$

$\boxed{B = -\frac{7}{16}}$

Hence the General sol

$a_n = \frac{11}{16}(1)^n - \frac{7}{16}(-1)^n - \frac{1}{4}\left(\frac{1}{2}\right)^n.$

(10) Solve the R.R $a_n - 7a_{n-1} + 10a_{n-2} = 0$ for $n \geq 2$ given that $a_0 = 10$ and $a_1 = 14$.

Sol:- Given $a_n - 7a_{n-1} + 10a_{n-2} = 0$, $n \geq 2$

The char equ $a^n - 7a^{n-1} + 10a^{n-2} = 0$

$$\begin{pmatrix} 1 & 0 \\ -7 & -2 \\ 0 & -7 \end{pmatrix}$$

Put $n=2$, $a^2 - 7a + 10 = 0$.

$$(a-5)(a-2) = 0$$

$$\boxed{a=5}, \boxed{a=2}$$

The char roots

$$a_n = A K_1^n + B K_2^n = A(5)^n + B(2)^n$$

Put $n=0$, $a_0 = A(5)^0 + B(2)^0 = A + B$

$$10 = A + B \rightarrow (1)$$

Put $n=1$, $a_1 = A(5)^1 + B(2)^1$

$$14 = 5A + 2B \rightarrow (2)$$

Solve (1) & (2)

$$(1) \times 2 \Rightarrow 2A + 2B = 20$$

$$(2) \times 1 \Rightarrow \begin{array}{r} 5A + 2B = 14 \\ (-) \quad (-) \quad (-) \\ \hline -3A = 6 \end{array}$$

$$\boxed{A = -2}$$

(1) $\Rightarrow$

$$A + B = 10$$

$$-2 + B = 10$$

$$B = 10 + 2$$

$$\boxed{B = 12}$$

The General sol is

$$a_n = (-2)(5)^n + 12(2)^n$$

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(11) Solve the R.R $a_n = 6a_{n-1} - 12a_{n-2} + 8a_{n-3}$ given that $a_0 = 1$, $a_1 = 4$, $a_2 = 28$.

Sol:- Given $a_n = 6a_{n-1} - 12a_{n-2} + 8a_{n-3}$.

We can write the above equ

$$a^n - 6a^{n-1} + 12a^{n-2} - 8a^{n-3} = 0$$

Put $n=3$, $a^3 - 6a^2 + 12a - 8 = 0$.

$\therefore a = 2$ is a root.

$$a^2 - 4a + 4 = 0$$

$$2 \left| \begin{array}{cccc} 1 & -6 & 12 & -8 \\ 0 & 2 & -8 & 8 \\ \hline 1 & -4 & 4 & 0 \end{array} \right|$$

$$(a-2)(a-2)=0$$

$$a=2, a=2.$$

$$\therefore a=2, 2, 2.$$

The char roots are 2, 2, 2

$$\therefore a_n = (A + Bn + Cn^2)K^n = (A + Bn + Cn^2)2^n.$$

Put $n=0$, $a_0 = (A + B(0) + C(0))2^0$

$$\boxed{1 = A}$$

Put $n=1$, $a_1 = (A + B + C)2^1$

$$4 = (1 + B + C)2$$

$$2 = 1 + B + C$$

$$1 = B + C \rightarrow \textcircled{1}$$

Put $n=2$, $a_2 = (A + 2B + 4C)2^2$

$$28 = 4(A + 2B + 4C)$$

$$7 = A + 2B + 4C$$

$$7 - 1 = 2B + 4C$$

$$6 = 2B + 4C$$

$$3 = B + 2C \rightarrow \textcircled{2}$$

Solve (1) & (2)

$$\textcircled{1} \Rightarrow B + C = 1$$

$$B + 2C = 3$$

$$\begin{array}{r} (-) \quad (-) \quad (-) \\ B + C = 1 \\ B + 2C = 3 \\ \hline -C = -2 \end{array}$$

$$-C = -2$$

$$\boxed{C = 2}$$

$$\textcircled{1} \Rightarrow 1 = B + C$$

$$1 = B + 2$$

$$\boxed{-1 = B}$$

Hence the solution is

$$a_n = (1 - n + 2n^2)2^n.$$

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Solution of Inhomogeneous Recurrence relations.

The eqn of the form $C_n a_n + C_{n-1} a_{n-1} + \dots + C_{n-k} a_{n-k} = f(n) \rightarrow (1)$

where $C_n, C_{n-1}, \dots, C_{n-k}$ are constants with $C_n \neq 0$ and $f(n)$ is a given real-valued fn of n .

The General solution of the R.R (1) is given by

$$a_n = a_n^{(h)} + a_n^{(p)} \rightarrow (2)$$

particular roots.



↘ particular solution.

To find P.S

If the R.H.S of the given R.R is a constant then replace $s(k), s(k-1), s(k-2)$ by d_0 (or) $s(k), s(k+1), s(k+2)$ by d_0 .

① solve the R.R $a_n + 4a_{n-1} + 4a_{n-2} = 8$ for $n \geq 2$ given that $a_0 = 1$ and $a_1 = 2$. (11)

Sol: Given $a_n + 4a_{n-1} + 4a_{n-2} = 8$

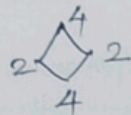
we can write the above eqn $a^n + 4a^{n-1} + 4a^{n-2} = 8$.

The char eqn

$$\text{put } n=2, a^2 + 4a + 4 = 0$$

$$(a+2)(a+2) = 0$$

$$a = -2, a = -2.$$



The char roots $a_n = (A+Bn)K^n = (A+Bn)(-2)^n$.

To find Particular solution.

The R.H.S of the R.R is constant.

Take $a_n = d_0, a_{n-1} = d_0, a_{n-2} = d_0$.

$$a_n + 4a_{n-1} + 4a_{n-2} = 8$$

$$d_0 + 4d_0 + 4d_0 = 8$$

$$9d_0 = 8$$

$$\boxed{d_0 = \frac{8}{9}}$$

The General solution is $a_n = a_n^{(h)} + a_n^{(p)}$

$$a_n = (A+Bn)(-2)^n + \frac{8}{9}$$

Given $a_0 = 1, a_1 = 2$.

$$\text{put } n=0, a_0 = (A+B(0))(-2)^0 + \frac{8}{9}$$

$$1 = A + \frac{8}{9}$$

$$1 - \frac{8}{9} = A$$

$$\boxed{\frac{1}{9} = A}$$

$$\text{put } n=1, a_1 = (A+B)(-2)^1 + \frac{8}{9}$$

$$2 = \left(\frac{1}{9} + B\right)(-2) + \frac{8}{9}$$

$$1 = -\frac{1}{9} - B + \frac{4}{9}$$

$$1 = \frac{3}{9} - B$$

$$1 - \frac{3}{9} = -B$$

$$\frac{6}{9} = -B$$

$$\boxed{B = -\frac{2}{3}}$$

Hence the solution is

$$a_n = \left(\frac{1}{9} - \frac{2}{3}n\right)(-2)^n + \frac{8}{9}$$

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To find Particular solution.

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The R.H.S is $c\lambda^k$ and $\lambda = a$ coincides with a characteristic root. when a is a simple root of the characteristic equation take $a_n^{(p)} = d_0 n$, & so on.

① Solve the recurrence relation $a_{n+2} - 4a_{n+1} + 3a_n = -200$, $n \geq 0$
and $a_0 = 3000$, $a_1 = 3300$.

Sol: Given $a_{n+2} - 4a_{n+1} + 3a_n = -200$ (✓) [$\because a^0 = 1$]

The char eqy

$$a^{n+2} - 4a^{n+1} + 3a^n = 0$$

$$\begin{array}{r} (x) \quad 3 \\ -3 \quad \swarrow -1 \\ \hline (x) \quad -4 \end{array}$$

Put $n=0$, $a^2 - 4a + 3 = 0$

$$(a-3)(a-1) = 0$$

$$a = 3, a = 1$$

The char roots $a_n^{(h)} = A K_1^n + B K_2^n = A(3)^n + B(1)^n$

Since 1 is a simple root of the char eqy and L.H.S of the given relation is a polynomial of degree 0, we seek $a_n^{(p)}$ in the form

$$a_n^{(p)} = d_0 n \rightarrow (1)$$

$$a_{n+1}^{(p)} = d_0 (n+1)$$

$$a_{n+2}^{(p)} = d_0 (n+2)$$

Given $a_{n+2} - 4a_{n+1} + 3a_n = -200$

$$d_0(n+2) - 4d_0(n+1) + 3d_0 n = -200$$

$$d_0 n + 2d_0 - 4d_0 n - 4d_0 + 3d_0 n = -200$$

$$-2d_0 = -200$$

$$\boxed{d_0 = 100}$$

Substituting $d_0 = 100$ in (1) we get

$$a_n^{(p)} = 100n$$

$\therefore$ The General soln is

$$a_n = 3A + B + 100n \rightarrow (2)$$

Given $a_0 = 3000$, $a_1 = 3300$

Put $n=0$ in (2)

$$a_0 = A + B(3)^0 + 100(0)$$

$$3000 = A + B \rightarrow (3)$$

Put $n=1$ in (2)

$$a_1 = 3A + B + 100$$

$$3300 = 3A + B + 100$$

$$3200 = 3A + B \rightarrow (4)$$

Solve (3) & (4)

$$A + B = 3000$$

$$3A + B = 3200$$

$$\begin{array}{r} (-) \quad (-) \quad (-) \\ \hline -2A = -200 \end{array}$$

$$\boxed{A = 100}, \boxed{B = 2900}$$

The General soln is

$$a_n = \underbrace{100(3)^n + 2900(1)^n}_{\alpha} + 100n$$

To find Particular solution.

(13)

Working Rule

If the R.H.S is $a_0 + a_1 k$ replace $S(k)$ by $d_0 + d_1 k$.

$$S(k) = d_0 + d_1 k.$$

$$S(k-1) = d_0 + d_1(k-1)$$

$$S(k-2) = d_0 + d_1(k-2) \dots \text{etc.}$$

$$S(k+1) \text{ by } d_0 + d_1(k+1)$$

$$S(k+2) \text{ by } d_0 + d_1(k+2) \dots \text{etc.}$$

① solve the R.R $S(k) - 7S(k-1) + 10S(k-2) = 6 + 8k$ with $S(0)=1$,
 $S(1)=2$.

Sol:- Given $S(k) - 7S(k-1) + 10S(k-2) = 6 + 8k$.

The char eqn

$$S(k) - 7S(k-1) + 10S(k-2) = 0.$$

$$a^k - 7a^{k-1} + 10a^{k-2} = 0.$$

$$\text{Put } k=2, a^2 - 7a + 10 = 0$$

$$(a-5)(a-2) = 0$$

$$a=5, a=2.$$

The char roots $a_n^{(k)} = A(k_1)^n + B(k_2)^n = A(5)^n + B(2)^n$.

To find Particular solution

As the R.H.S of the given relation is $6 + 8k$

Take $S(k) = d_0 + d_1 k$, $S(k-1) = d_0 + d_1(k-1)$, $S(k-2) = d_0 + d_1(k-2)$ respectively.

$$S(k) - 7S(k-1) + 10S(k-2) = 6 + 8k$$

$$[d_0 + d_1 k] - 7[d_0 + d_1(k-1)] + 10[d_0 + d_1(k-2)] = 6 + 8k$$

$$4d_0 - 13d_1 + 4d_1 k = 6 + 8k$$

Equating like the coefficients $4d_1 = 8 \Rightarrow \boxed{d_1 = 2}$

$$4d_0 - 13d_1 = 6$$

$$4d_0 - 13(2) = 6$$

$$4d_0 = 6 + 26 = 32 \Rightarrow \boxed{d_0 = 8}$$

Hence the General solution is

$$a_n = A(5)^n + B(2)^n + 8 + 2n$$

Put $n=0$, $a_0 = A(5)^0 + B(2)^0 + 8 + 2(0)$

$$1 = A + B + 8$$

$$A + B = -7 \rightarrow (1)$$

Put $n=1$, $a_1 = A(5)^1 + B(2)^1 + 8 + 2(1)$

$$2 = 5A + 2B + 8 + 2$$

$$-8 = 5A + 2B \rightarrow (2)$$

Solve (1) & (2)

$$(1) \times 2 \Rightarrow 2A + 2B = -14$$

$$(2) \times 1 \Rightarrow \begin{array}{r} 5A + 2B = -8 \\ (-) \quad (-) \quad (+) \\ \hline -3A = -6 \end{array}$$

$$A = 2$$

$$(1) \Rightarrow A + B = -7$$

$$2 + B = -7$$

$$B = -7 - 2 = -9$$

$$B = -9$$

Hence the general solution is $a_n = 2(5)^n - 9(2)^n + 8 + 2n$.

② Solve the recurrence relation $a_{n+2} - 7a_{n+1} + 10a_n = 8n + 6$.

Sol:- We can write the eqn $a^{n+2} - 7a^{n+1} + 10a^n = 8n + 6$.

The char eqn $a^{n+2} - 7a^{n+1} + 10a^n = 0$.

Put $n=0$, $a^2 - 7a + 10 = 0$

$$\begin{array}{c} 10 \\ -5 \quad \swarrow \quad \searrow \\ -7 \end{array} -2$$

$$(a-5)(a-2) = 0.$$

$$a = 5, a = 2.$$

The char roots $a_n^{(h)} = A(K_1)^n + B(K_2)^n = A(5)^n + B(2)^n$.

To find Particular solution

Take $a_n = d_0 + d_1 n$, $a_{n+1} = d_0 + d_1(n+1)$, $a_{n+2} = d_0 + d_1(n+2)$
 Now

$$a_{n+2} - 7a_{n+1} + 10a_n = 8n + 6$$

$$[d_0 + d_1(n+2)] - 7[d_0 + d_1(n+1)] + 10[d_0 + d_1 n] = 8n + 6$$

$$d_0 + d_1(n+2) - 7d_0 - 7d_1(n+1) + 10d_0 + 10d_1n = 8n+6$$

$$\underline{d_0} + d_1n + \underline{2d_1} - \underline{7d_0} - 7d_1n - \underline{7d_1} + \underline{10d_0} + 10d_1n = 8n+6$$

$$4d_0 - 5d_1 + 4d_1n = 8n+6$$

Equating like the coefficients

$$\begin{array}{l|l|l} 4d_1 = 8 & 4d_0 - 5d_1 = 6 & 4d_0 = 16 \\ \boxed{d_1 = 2} & 4d_0 - 5(2) = 6 & \boxed{d_0 = 4} \\ & 4d_0 - 10 = 6 & \end{array}$$

Hence the General solution is $a_n^{(H)} = A(5)^n + B(2)^n + d_0 + d_1n$
 $= A(5)^n + B(2)^n + 4 + 2n.$

Particular solution

Working Rule:

Suppose $f(n) = ab^n$ where a is a constant and b is not a root of the char eqn of the homogeneous part of the relation. Then $a_n^{(P)}$ is taken in the form $a_n^{(P)} = ab^n$.

Q Solve the recurrence relation $a_{n+2} + 3a_{n+1} + 2a_n = 3^n$ for $n \geq 0$
 given $a_0 = 0, a_1 = 1$.

Soln: The char eqn is $a^{n+2} + 3a^{n+1} + 2a^n = 0$

Put $n=0$, $a^2 + 3a + 2 = 0$

$$(a+2)(a+1) = 0.$$

$$a_1 = -2, a_2 = -1.$$

The char roots of $a_n^{(H)} = A(K_1)^n + B(K_2)^n = A(-2)^n + B(-1)^n$.

The RHS of given eqn is 3^n .

$$a_n^{(P)} = d_0 3^n \text{ (lets form } d_0 b^n)$$

$$\text{Take } a_{n+1} = d_0 3^{n+1}$$

$$a_{n+2} = d_0 3^{n+2}$$

Given $a_{n+2} + 3a_{n+1} + 2a_n = 3^n$

$$d_0 3^{n+2} + 3d_0 3^{n+1} + 2d_0 3^n = 3^n$$

$$3^n [d_0 9 + 9d_0 + 2d_0] = 3^n$$

$$K d_0 4^K - 3(d_0 4^{K-1}(K-1)) - 4(d_0 4^{K-2}(K-2)) = 4^K$$

$$2d_0 = 1$$

$$\boxed{d_0 = \frac{1}{2d_0}}$$

$$\begin{array}{c} \textcircled{x} \quad 2 \\ \textcircled{+} \quad 1 \\ \hline 3 \end{array}$$

$$\therefore a_n^{(P)} = \frac{1}{20} (2)^n$$

The general solution is $a_n = A(-2)^n + B(-1)^n + \frac{1}{20} (2)^n \Rightarrow (1)$

Given $a_0 = 0, a_1 = 1$

Put $n=0$ in (1) $a_0 = A + B + \frac{1}{20}$

$$0 = A + B + \frac{1}{20}$$

$$A + B = -\frac{1}{20} \Rightarrow (2)$$

Put $n=1$ in (1) $a_1 = A(-2)^1 + B(-1)^1 + \frac{1}{20} (2)^1$

$$1 = -2A - B + \frac{2}{20}$$

$$\frac{17}{20} = -2A - B \Rightarrow (3)$$

Solve (2) & (3)

$$A = -\frac{4}{5}$$

$$B = \frac{3}{4}$$

Hence the required a_n is

$$a_n = \left(-\frac{4}{5}\right)(-2)^n + \left(\frac{3}{4}\right)(-1)^n + \frac{1}{20} (2)^n$$

(2) Solve the R.R. $S(K) = 3S(K-1) - 4S(K-2) = \frac{1}{4}^K, K \geq 2$ using R.R.

Soln Given $a^K = 3a^{K-1} - 4a^{K-2} = \frac{1}{4}^K, K \geq 2$

The char eqn $a^K - 3a^{K-1} - 4a^{K-2} = 0$

$$= 4 \begin{matrix} -4 \\ 1 \\ -3 \end{matrix}$$

Put $K=2, a^2 - 3a - 4 = 0$

$$(a-4)(a+1) = 0$$

$$\boxed{a=4} \quad \boxed{a=-1}$$

The char roots $a_n = A(K_1)^n + B(K_2)^n = A(4)^n + B(-1)^n$

The R.H.S of given eqn is $\frac{1}{4}^K$. The base of R.H.S is 4. As 4 is a simple root of the char eqn

$$a_n^{(P)} = K d_0 4^K \quad (\text{ie, it is form } K d_0 h^n) \quad \text{eqn}$$

Take $a_{K-1} = d_0 4^{K-1} (K-1)$

$$a_{K-2} = d_0 4^{K-2} (K-2)$$

Given $a^K - 3a^{K-1} - 4a^{K-2} = \frac{1}{4}^K$

$$K d_0 4^K - 3(d_0 4^{K-1} (K-1)) - 4(d_0 4^{K-2} (K-2)) = \frac{1}{4}^K$$

$$4^k [d_0 k - 3d_0 4^{-1}(k-1) - 4d_0 4^{-2}(k-2)] = 4^k$$

$$d_0 k - \frac{3}{4} d_0 (k-1) - \frac{1}{4} d_0 (k-2) = 1$$

$$4d_0 k = 3d_0 (k-1) + d_0 (k-2) + 4$$

$$4d_0 k - 3d_0 k + 3d_0 = d_0 k + d_0 - 2d_0 + 4$$

$$5d_0 = 4$$

$$d_0 = \frac{4}{5}$$

The Particular solution is $a_n^{(p)} = d_0 k 4^n = \left(\frac{4}{5}\right) k 4^n$

Hence the General solution is $a_n = a_n^{(h)} + a_n^{(p)}$

$$= A(4)^n + B(-1)^n + \left(\frac{4}{5}\right) k 4^n$$

③ Solve the recurrence relation $a_n = 5a_{n-1} - 6a_{n-2} + 7^n$

sol: Given $a^n - 5a^{n-1} + 6a^{n-2} = 7^n$

The char eqn

Put $n=2$, $a^2 - 5a + 6 = 0$

$$(a-3)(a-2) = 0$$

$$a = 2, 3$$

The char roots $a_n = A(K_1)^n + B(K_2)^n = A(2)^n + B(3)^n$

The RHS of given eqn is 7^n

[i.e., the form $d_0 7^n$]

Take $a_n^{(p)} = d_0 7^n$

$$a_{n-1} = d_0 7^{n-1}$$

$$a_{n-2} = d_0 7^{n-2}$$

Given $a^n - 5a^{n-1} + 6a^{n-2} = 7^n$

$$d_0 7^n - 5d_0 7^{n-1} + 6d_0 7^{n-2} = 7^n$$

$$7^n \left[d_0 - \frac{5}{7} d_0 + \frac{6}{49} d_0 \right] = 7^n$$

$$d_0 - \frac{5}{7} d_0 + \frac{6}{49} d_0 = 1$$

$$49d_0 - 35d_0 + 6d_0 = 49$$

$$20d_0 = 49 \Rightarrow$$

$$d_0 = \frac{49}{20}$$

$$\therefore a_n^{(p)} = d_0 7^n$$

$$= \frac{49}{20} (7)^n$$

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The general solution is $a_n = A(2)^n + B(3)^n + \frac{49}{20} (7)^n$

① Solve the $S(k) - 4S(k-1) + 4S(k-2) = 3k + 2^k$, $k \geq 2$ with $S(0)=1$, $S(1)=1$.

Sol: The char eqn Given $S(k) - 4S(k-1) + 4S(k-2) = 3k + 2^k$

$$a^k - 4a^{k-1} + 4a^{k-2} = 0$$

$$\begin{array}{c} 4 \\ -2 \quad \triangle \quad -2 \\ -4 \end{array}$$

Put $k=2$, $a^2 - 4a + 4 = 0$

$$(a-2)(a-2) = 0$$

$$a = 2, \quad a = 2$$

The char roots $a_n^{(h)} = (A+Bn)K^n = (A+Bn)2^n$

To find particular solution for $3k$

Take $d_0 + d_1 k$ Replace $S(k) = d_0 + d_1 k$
 $S(k-1) = d_0 + d_1 (k-1)$
 $S(k-2) = d_0 + d_1 (k-2)$

Given $S(k) - 4S(k-1) + 4S(k-2) = 3k$

$$[d_0 + d_1 k] - 4[d_0 + d_1 (k-1)] + 4[d_0 + d_1 (k-2)] = 3k$$

$$[d_0 - 4d_0 + 4d_0] + [d_1 k - 4d_1 k + 4d_1 + 4d_1 k - 8d_1] = 3k$$

$$d_0 - 4d_1 + d_1 k = 3k$$

Equating like the coefficients $d_0 - 4d_1 = 0$ $\left| \begin{array}{l} d_1 = 3 \\ d_0 = 12 \end{array} \right.$

$$\therefore a_n^{(p)} = d_0 + d_1 k = 12 + 3k$$

To find particular solution for 2^k

Take $d k^2 2^k$ (Since the base of the RHS, i.e., 2 is a double root of the char eqn) Replace $S(k) = d k^2 2^k$
 $S(k-1) = d (k-1)^2 2^{k-1}$
 $S(k-2) = d (k-2)^2 2^{k-2}$

Given $s(k) - 4s(k-1) + 4s(k-2) = 2^k$

$$dk^2 2^k - 4d(k-1)^2 2^{k-1} + 4d(k-2)^2 2^{k-2} = 2^k$$

$$2^k \left[dk^2 - \frac{4}{2} d(k-1)^2 + \frac{4}{4} d(k-2)^2 \right] = 2^k$$

$$dk^2 - 2d(k-1)^2 + d(k-2)^2 = 1$$

$$dk^2 - 2d(k^2 + 1 - 2k) + d(k^2 + 4 - 4k) = 1$$

$$\underline{dk^2} - \underline{2dk^2} - 2d + 4dk + \underline{dk^2} + 4d - 4dk = 1 \quad (\times)$$

$$2d = 1$$

$$\boxed{d = \frac{1}{2}}$$

Particular solution $a_n^{(P)} = dk^2 2^k$
 $= \left(\frac{1}{2}\right) k^2 2^k$

Hence the General solution is

$$a_n = s(n) = s(k) = (A + Bn)2^n + 12 + 3n + \frac{1}{2} n^2 2^n \quad (k=n)$$

Given $s(0) = 1$,

$s(1) = 1$.

Put $k=0$,

$$s(0) = (A + 0) + 12 + 0 + 0$$

$$1 = A + 12$$

$$\boxed{A = -11}$$

Put $n=1$,

$$s(1) = (A + B)2 + 12 + 3 + \frac{1}{2}(1)^2 2^1$$

$$1 = 2(A + B) + 15 + 1$$

$$1 - 16 = 2A + 2B$$

$$-15 = 2A + 2B$$

$$-15 - 2A = 2B$$

$$-15 - 2(-11) = 2B$$

$$-15 + 22 = 2B$$

$$7 = 2B \Rightarrow \boxed{B = \frac{7}{2}}$$

Hence the General solution is

$$a_n = \left(-11 + \frac{7}{2}n\right)2^n + 12 + 3n + \frac{1}{2} n^2 2^n$$

Solution of R.R. by using Z-Transform

(20)

Def: The generating fn. of a sequence $a_0, a_1, \dots, a_n, \dots$ is the expression

$$G(x) = a_0 + a_1x + a_2x^2 + \dots = \sum_{n=0}^{\infty} a_n x^n$$

(1) Use the method of generating fn. to solve the R.R. $a_n = 3a_{n-1} + 1$,

$n \geq 1$ given that $a_0 = 1$.

Sol: w.k.t. $G(x) = \sum_{n=0}^{\infty} a_n x^n$.

Given $a_n = 3a_{n-1} + 1$

Take $\sum_{n=1}^{\infty} 3a_{n-1}x^n$ both sides

$$\sum_{n=1}^{\infty} a_n x^n = 3 \sum_{n=1}^{\infty} a_{n-1} x^n + \sum_{n=1}^{\infty} (1)x^n$$

$$a_1x^1 + a_2x^2 + \dots = 3[a_0x^1 + a_1x^2 + a_2x^3 + \dots] + [x + x^2 + x^3 + \dots]$$

$$[a_0 + a_1x + a_2x^2 + \dots - a_0] = 3x[a_0 + a_1x + a_2x^2 + \dots] + x[1 + x + x^2 + \dots]$$

$$G(x) - a_0 = 3xG(x) + x(1-x)^{-1}$$

$$G(x) - 1 = 3xG(x) + x(1-x)^{-1}$$

$$(1-3x)G(x) = 1 + \frac{x}{(1-x)}$$

$$(1-3x)G(x) = \frac{1+x}{(1-x)}$$

$$G(x) = \frac{1}{(1-x)(1-3x)}$$

$$\text{Consider } \frac{1}{(1-x)(1-3x)} = \frac{A}{1-x} + \frac{B}{1-3x}$$

$$1 = A(1-3x) + B(1-x)$$

$$1 = 0 + B(1-\frac{1}{3})$$

$$\text{Put } x = \frac{1}{3},$$

$$1 = \frac{2}{3}B$$

$$\boxed{\frac{3}{2} = B}$$

$$1 = A(1-3) + 0$$

$$\text{Put } x = 1,$$

$$1 = -2A$$

$$\boxed{-\frac{1}{2} = A}$$

$$G(x) = \frac{-1/2}{1-x} + \frac{3/2}{1-3x}$$

$$= -\frac{1}{2}(1-x)^{-1} + \frac{3}{2}(1-3x)^{-1}$$

$$= -\frac{1}{2}(1+x+x^2+\dots+x^n+\dots)$$

$$+ \frac{3}{2}(1+(3x)+(3x)^2+\dots+(3x)^n+\dots)$$

The coefficient of x^n is

$$G(x) = -\frac{1}{2}(1)^n + \frac{3}{2}(3)^n$$

$$= -\frac{1}{2} + \frac{3^{n+1}}{2}$$

$$\underline{\underline{-\frac{1}{2} + \frac{3^{n+1}}{2}}}$$